

RITUPARNA WARWATKAR

Portfolio ☎ +91 77985 16764 ✉ hello@rituu.com
in rituparna-warwatkar ○ rituparna-ui 📁 rituparna-warwatkar

EDUCATION

Indian Institute of Information Technology, Guwahati

Dec 2020 – May 2024

Bachelor of Technology, Computer Science and Engineering

CGPA: 8.21

TECHNICAL SKILLS

Languages: C, aarch64, TypeScript, Python, Dart, Java

Database: MongoDB, Redis, SQL, Firebase, Pinecone

Technologies: Linux, KVM, Docker, ME(F/A/R/V)N Stack

Other: AWS, Git, Shell Scripting

EXPERIENCE

Amazon Web Services - EC2

Feb 2025 – Present

Kernel/Hypervisor Engineer

Berlin, Germany

Linux | KVM | EC2 Nitro | Virtualization | PCIe

- Investigated **platform hardware failures** by analyzing large-scale logs and debugging, proposing remediation strategies that improved fleet health and **reduced** unsellable capacity by **~60%**.
- Contributed to **bring-up** and platform enablement of next-generation accelerated platforms (**P6 B200/B300**), involving configuration of memory maps, PCIe ACS and enabling reliable **GPU passthrough** for customer workloads.
- Implemented and enabled NVIDIA **GPUDirect RDMA** and **GPUDirect Storage** on P6 platforms, allowing direct data transfer between GPUs, NICs, and storage without CPU intervention.
- Drove end-to-end global **deployment** and rigorous testing for the **hypervisor** and critical Nitro components, ensuring stability and security across a variety of Accelerated Computing platforms.
- Conducted in-depth **benchmarking** and stress testing across accelerated platforms and debugging failures to guarantee peak performance and critical stability in production.
- Built** and **automated** various testing payloads including **GPU Fuzzing**, Deep Learning training, LLM inferencing and Graphics/rendering workloads to validate system performance and stability for both GPUs and CPUs.
- Managed critical production **alerts** during **on-call** rotation, quickly diagnosing and debugging failures to **minimize downtime** and prevent recurrence.

Veritus.ai

May 2024 – Sept 2024

Software Development Engineer

Kobe, Japan

Node.js | MongoDB | Redis | React | AWS | Next.js | tRPC | OpenAI | Gemini | Pinecone

- Implemented a comprehensive authentication system, including user signup, waitlist management, login, **OAuth social login**, email verification, and password management.
- Developed and integrated a robust **payment processing** and credit control system using **Stripe**.
- Developed a fully-featured full-stack **admin panel** for managing users and platform content, including features such as waitlist management, user transactions, and user feedback.
- Developed a **feature flag** system to enable or disable features without modifying source code or redeploying, allowing for **controlled feature rollouts** and **targeted user testing**.
- Implemented context caching to optimize AI workflows, **reducing LLM costs** by up to **65%** by reusing cached tokens instead of passing the same input repeatedly.
- Implemented a LaTeX renderer for rendering beautiful mathematical expressions and equations generated by the LLMs.

FIITJEE, Edfora

Dec 2023 – April 2024

Software Developer Intern

Delhi, India

Node.js | MongoDB | Redis | Angular | React | Swagger

- Developed a full-stack **dynamic form builder** and response system, enabling institute admins to create and assign custom forms to students, enhancing platform engagement and **eliminating reliance** on third-party services like Google Forms.
- Implemented **device registration and management** for user authentication on the Glorifire Jr. platform, utilizing **fingerprint.js** for the web and **device IDs** for mobile app. tm
- Developed a full-stack "Question Contribution" module, enabling students to contribute **various question types**, including multiple choice, numerical, and matrix match, to the question bank.
- Enhanced student engagement by up to **50%** through push notifications, emails, and in-app notifications.

PROJECTS

Fermi OS | ○

C, ARMv8-A, QEMU, VirtIO over PCIe

- A preemptive multi-tasking AArch64 kernel with **EL0/EL1** privilege separation, 3-level **MMU/page tables**, per-process **virtual address spaces**, kernel heap, physical memory manager, and ARM **exception/interrupt** handling.
- POSIX**-style syscall and userspace with *svc* dispatch, **process scheduler**, file descriptors, VFS, **FAT32** and */proc*.
- Developed **PCIe subsystem** and **VirtIO** over PCIe drivers for block, network, and RNG devices.
- Implemented **GICv3** interrupt handling, ARM Generic Timer-based preemption, task sleeping/wakeup, **context switching**, and kernel fault isolation; see [GitHub README](#) for additional features and implementation details.